

Systematic Literature Review Protocol

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The impacts of artificial intelligence on environmental sustainability and human well-being	
Requirement	Description
Rationale	The development of artificial intelligence (AI) is already having a significant impact on society and the environment, and this impact is predicted to increase dramatically in the coming years. The aim of this systematic literature review is to bring together contributions relating to the impacts of AI on environmental sustainability and human well-being, analyse how this literature is composed in terms of impact types considered and methods used, and identify key findings and research gaps to guide future work. This review is intended to be the basis for further work to quantitatively and qualitatively assess the impacts of AI on environmental sustainability and human well-being under different scenarios of AI development.
Objectives	- Explore the impacts that AI development could have on environmental sustainability. - Explore the impacts that AI development could have on human well-being. - Identify the methods used in the literature to quantitatively and qualitatively assess the impacts of AI on environmental sustainability and human well-being. - Identify research gaps and priorities for future work to promote a comprehensive understanding of the impacts of AI on environmental sustainability and human well-being.
Strategy	- Initial search in identified databases (Scopus, arXiv, NBER working papers). - Complement the initial search with forward and backward citation tracking to identify additional relevant studies, including screening the reference lists of included papers (backward citation tracking) and identifying subsequent studies that cite them (forward citation tracking).
Research questions	Q1: What impacts does the literature suggest that AI could have on the environment? Q2: What impacts does the literature suggest that AI could have on human well-being? Q3: What methods are used in the literature to assess the environmental and human well-being impacts of AI? Q4: What research gaps and priorities for future work emerge?
Keywords	Environment: Artificial Intelligence, AI, Machine Learning, ML, Deep Learning, Large Language Models, LLMs, environment, environmental, sustainability, sustainable, carbon, biodiversity, ecological, climate change, energy, resource, impact, effect, footprint Human well-being: Artificial Intelligence, AI, Machine Learning, ML, Deep Learning, Large Language Models, LLMs, wellbeing, well-being, happiness, health, employment, unemployment, job, labour, labor, equality, inequality, society, impact, effect, implication
Sources	- Scopus - ArXiv - NBER working papers
Type of search	Title, abstract, and keywords
Timeframe	2010 to 2024

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Geographical focus	Global (English documentation)
Search strings & results	For each thematic group (environment, human well-being), we developed a specific search string. The search strings were then adapted to the syntax and functionality of each database. The process began with Scopus, which offers the most advanced search tools, and the process was subsequently adapted for arXiv and the NBER Working Papers (for which the advanced search was carried out through Google Scholar indexing). Scopus: ➤ Environmental impacts: TITLE-ABS-KEY ((impact* OR footprint* OR effect) W/3 (ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm* OR "Large Language Model*") W/3 (environment* OR sustainab* OR ecological OR biodiversity OR carbon OR "climate change" OR energy OR resource*)) AND PUBYEAR > 2009 AND PUBYEAR < 2025 AND (LIMIT-TO (LANGUAGE , "English")) Hits: 1414 (Last access 31/12/2024) Notes: The different positioning possibilities of the 3 keyword groups were considered and having both the impact-related keyword group and the environment-related keyword group close to the AI-related keyword group returned more results, encompassing the largest number of phrases relevant to the research question. ➤ Human well-being impacts: TITLE-ABS-KEY ((impact* OR implication* OR effect) W/3 (ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm* OR "Large Language Model*") W/3 (wellbeing OR well-being OR happiness OR health OR *employment OR labor OR labour OR job* OR *equality OR society)) AND PUBYEAR > 2009 AND PUBYEAR < 2025 AND (LIMIT-TO (LANGUAGE,"English")) Hits: 1449 (Last access 31/12/2024) Notes: Alternative words referring to AI social and well-being impacts were tested such as life satisfaction, time use, work, equity, community, and social cohesion. The most complete combination of words yielding relevant results was finally selected. Since the other databases do not support proximity operators, the operator W/ was replaced by AND. To restrict the search to articles discussing impacts of artificial intelligence, the AI-related keyword group was searched in the title. arXiv ➤ Environmental impacts: date_range: from 2010-01-01 to 2024-12-31; AND title=ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm OR "Large Language Model"; AND abstract=environment OR environmental OR sustainable OR sustainability OR ecological OR biodiversity OR carbon OR "climate change" OR energy OR resources; AND abstract=impact OR footprint Hits: 1412 (Last access 31/12/2024) 1342 retrieved through arXiv API Notes: The word “effect” was excluded because the arXiv search engine does not strictly enforce exact phrase matching resulting in unwanted matches such as “effectiveness”, “effectively”, or “effective” which are very common words and cause many irrelevant results to be included in the list.

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	➤ Human well-being impacts: date_range: from 2010-01-01 to 2024-12-31; AND title=ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm OR "Large Language Model"; AND abstract=wellbeing OR well-being OR happiness OR health OR employment OR unemployment OR labor OR labour OR job OR equality OR inequality OR society; AND abstract=impact OR implication Hits: 1072 (Last access 31/12/2024) 927 retrieved through arXiv API NBER working papers (via Google Scholar indexing) ➤ Environmental impacts: site:nber.org (intitle:(ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm OR "Large Language Model")) AND(impact OR footprint OR effect) AND (environment OR environmental OR sustainable OR sustainability OR ecological OR biodiversity OR carbon OR "climate change" OR energy OR resources) Date: 2010-2024 Hits: 948 (Last access 31/12/2024) 652 retrieved through Zotero ➤ Human well-being impacts: site:nber.org (intitle:(ai OR "Artificial Intelligence" OR ml OR "Machine Learning" OR "Deep Learning" OR llm OR "Large Language Model")) AND (impact OR effect OR implications) AND (wellbeing OR well-being OR happiness OR health OR employment OR unemployment OR labor OR labour OR job OR equality OR inequality OR society) Date: 2010-2024 Hits: 930 (Last access 31/12/2024) 646 retrieved through Zotero Notes: The output records (948 and 930 respectively) included many papers published before 2010, the number of retrieved studies reported is net of deleted records falling outside the 2010-2024 period.
Eligibility criteria	• Published on or after 2010 • Peer-reviewed articles, conference proceedings, working papers, preprints, or grey literature from reputable institutions • In English language • Full text available • Relevance to the research questions
Data management	• The full electronic search strategy for the major databases, including any limits applied, is documented in the systematic review protocol such that it could be repeated. • All information sources in the search are documented in the systematic review protocol and PRISMA Flow Diagram. • The exact number of references retrieved, screened, selected, included, and excluded from the review is documented in the PRISMA Flow Diagram. • Bibliographic references were managed using Zotero. • Screening and data extraction were carried out using structured Excel spreadsheets. • All review files were stored in a shared folder on Teams to ensure secure access and version control within the project team.
Selection process	• Author/s, Title, Year, Journal, Link, and Abstract of retrieved studies were downloaded in a spreadsheet. • A second version of the spreadsheet containing only the titles and abstracts of the studies was created for the authors to proceed with the title and abstract screening blind to authorship and publication source. • Prior to first screening, duplicates were removed. • After first screening, full-text studies were checked for eligibility, providing inclusion/exclusion reasons and explanations in the PRISMA Flow Diagram.

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	Available and eligible studies were included in the review and classified by relevance to the research questions into three categories: very relevant (the study's core contribution directly addresses one or more research questions), relevant (the study addresses one or more research questions, but this is not its core contribution), and less relevant (the study tangentially addresses or intersects with one or more research questions).
Data extraction	Data were extracted from studies through a data extraction table structured in three main sections: - A first section reporting bibliographic information (author/s, title, year, journal, link/DOI) and a brief summary of the studies synthesising their aim, method and key findings. - A second section including relevant categories to which studies might pertain with drop-down menus or checkboxes (research line, scale of analysis, broad method, specific method, impact type, level at which impacts arise, and sentiment). Categories were developed iteratively during an initial piloting phase and refined throughout the review process. - A third section with open-text fields reporting the key elements of the studies: key findings, methods, limitations, research gaps and notes. The categories included in the second section are as follows: - Research line: AI environmental impact, AI well-being impact. Studies could be cross-coded if they addressed both. - Scale of analysis: macro (global, cross-country); macro (national, sub-national); micro (enterprise; set of projects/models); micro (application/model/hardware). - Broad method: conceptual/theoretical, qualitative, or quantitative. If a study used multiple methods, it was categorised according to the predominant method or the method that had the greatest influence on the results. - Specific method: - Conceptual/theoretical: Conceptual Framework, Scoping/ Literature Review, Conceptual System Design, Ethical Analysis, Policy Analysis, Perspective Analysis. - Qualitative: Qualitative Experimental Evaluation, Interview, Survey, Case Study, Participatory Method, Thematic/ Content Analysis, Expert Elicitation. - Quantitative: Quantitative Experimental Evaluation, Quantitative Survey, Quantitative Case Study, Statistical Analysis, Macroeconomic Analysis, Life Cycle Assessment, Optimisation, Remote Sensing. - Impact type: - Environmental: energy use, material use, water use, land use, ecological footprint, CO₂ emissions, other pollutants, biodiversity, sustainable development goals (SDGs). - Well-being: income, employment, inequality, health, subjective well-being, cognitive capabilities, social cohesion, existential risk. Studies could be cross-coded if they addressed multiple impacts. - Level at which impacts arise (following the framework proposed by Kaack et al., 2022): computing-related, application-level, or systemic. For environmental studies at the application and systemic level we also coded the most recurring impact pathways: - Application-level: improvements in firms' environmental performance, or other. - Systemic: rebound effects, shifts in consumer behaviour, or other. - Sentiment: positive, negative, or neutral. The sentiment was assigned at the study level based on the authors' own assessment of the impact. If a study reported only benefits and no harm, it was classified as positive. Conversely, if it reported only harm and no benefits, it was classified as negative. If both aspects were reported without a clear overall trend, or if the description was

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	explicitly ambivalent, the study was classified as neutral. Where both benefits and harms were reported, but the authors indicated an overall trend, for example in the abstract or discussion, we followed that indication. For studies classified as very relevant and relevant, data extraction was performed manually. For studies classified as less relevant, data extraction was supported by a Large Language Model (LLM), specifically Microsoft 365 Copilot Enterprise. LLM-assisted data extraction was performed using two standardised prompts. Each prompt was developed iteratively, monitoring the quality of the output until the expected result was achieved. Before including the extracted information in the data extraction spreadsheet, each individual datum was verified against the content of the original study. Outputs of data extraction were: ○ study summary to get an overview of the studies' content; ○ categorisation of the publications into research line, scale of analysis, broad method, specific method, impact type, level at which impacts arise, and sentiment categories; ○ identification of the key findings of the study; ○ identification of the methods employed by the study and associated limitations; ○ identification of the main research gaps highlighted by the study.
Quality assessment	• At the review level, quality was ensured via the application of the PRISMA 2020 framework, checklist and guidelines. • Risk of bias of individual studies was assessed at study level during full-text screening via a quality checklist (data support, causal links, assumptions disclosure, proxies and measurements validation, circular reasoning, selective reporting). • Studies classified as at critical risk of bias were excluded from the review and the reason is reported in the PRISMA Flow Diagram. • For LLM-assisted extracted data the quality control checklist included an item for each piece of extracted information, verifying its accuracy and cross-checking it against the original text. • Spot checks were carried out on category tags on a random subset of records, cross-checking data with the source PDFs.
Synthesis	The review includes both quantitative and qualitative analysis. • The quantitative analysis maps the distribution of the studies across the different categories (research line, scale of analysis, broad method, specific method, impact type, level at which impacts arise, and sentiment), highlighting literature focus and framing, and identifying research gaps. • The qualitative analysis identifies: ○ recurring and emerging findings and "key findings", defined as findings characterised by higher impact severity and a higher degree of confidence; ○ priorities for future work, identified from both the research gaps from the quantitative analysis and the research gaps reported by the studies included in the review.
Contributions	• Protocol and review author: Noemi Luna Carmeno • Protocol and review author: Tiago Domingos • Protocol and review guarantor: Daniel W. O'Neill
Amendments	Any amendments to the protocol are validated by the review guarantor, justified and recorded in the updated protocol.